

Experiment Title :-

Study of LAN Topology, Network Devices, and Address Configuration.

Aim :-

To study the LAN topology used in the organization, identify networking devices analyze wired and wireless connections measure network speed and determine system network parameters such as MAC Address, IP Address, and Subnet Mask.

Objectives :-

- 1) To understand LAN topology and networking components.
- 2) To identify the role of networking devices.
- 3) To analyze wired and wireless network connections.
- 4) To measure network ~~for~~ speed.
- 5) To determine MAC address, IPV4 Address and subnet Mask.
- 6) To understand the role of subnet masking in networking.

Theory :-

A Local Area Network (LAN) is a network that connects computers and devices within a limited geographical area such as a home, office, or college campus. LAN provides resource sharing internet access and communication between connected devices.

Network topology defines the arrangement of devices in a network. Common topologies includes Star, Bus, Ring and Mesh topology. Modern LANs mostly use star topology due to better performance and easier maintenance.

Networking devices such as routers, switches and access points help in communication and data transfer between devices.

Problem Statement:-

1. Draw the Network Topology

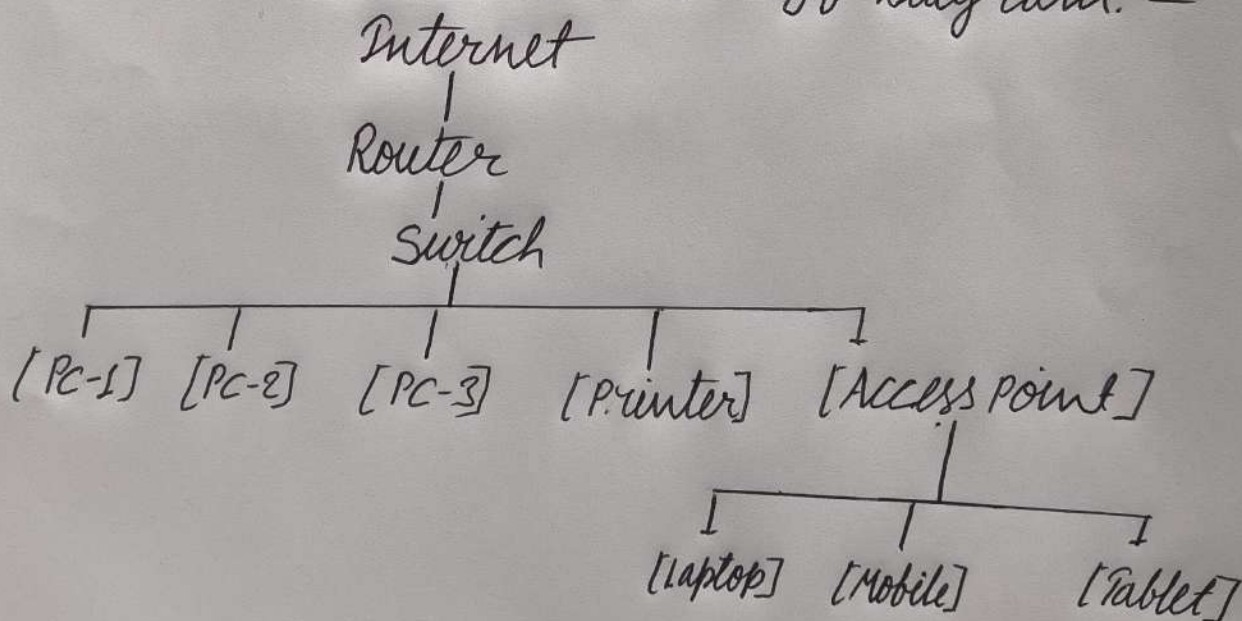
- Identify and draw the LAN topology of your organisation (eg star, Bus, Mesh).
- clearly label all components such as Router, switch, Access points and end devices (PCs, laptops etc.)

Identified LAN Topology: star Topology

The LAN Topology used in the organization is star topology. In this topology, all devices are connected to a central devices called a switch or Router. Communication between devices takes place through the central devices.

star topology is widely used in school, colleges offices and homes because it provides better performance easy management and reliability.

~~Network~~ Network Topology Diagram:-



Components in Topology

Component

Router

Switch

Access point

PC-1 / PC-2

Printer

Laptop

Mobile / Table

Function

Connects the LAN to the internet
Central device connecting wired system.

Provides wireless wi-fi connectivity
Desktop computers in the LAN

Shared network printer

wireless client device

wifi enable smart devices.

2. Identify Network Devices:-

- List all networking devices used in the LAN.
- Mention the function of each device.

Network Devices used in LAN:-

In a Local Area Network (LAN) different networking devices are used to connect computers, manage data flow and provide internet access.

List of Network Devices:-

1. Router:-

Function:-

- connects the LAN to the Internet (WAN)
- Routes data packets between network
- Assigns IP addresses using DHCP (in most case)
- Provides security feature like NAT and Firewall.

2. Switch:-

Function:-

- connects multiple devices within the LAN
- Transfers data only to the intended device using MAC address.
- Reduces network traffic and improves performance
- Act as the central device in star topology

3. Access Point (AP)

Function:-

- Provides wireless connectivity (Wi-Fi)
- connects wireless devices like laptop, mobiles, tablets to the LAN
- Extends network coverage without cables

4. Modem (if used)

Function:-

- Converts digital signals to analog and vice versa.
- Connects the internal network to the ISP (Internet service provider).
- Works as an interface between LAN and Internet line.

5. Ethernet cable

Function:-

- Provides wired connection between devices.
- Ensure high-speed and stable data transmission.
- Used to connect PCs, switches, and routers.

6. Network Interface card (NIC)

Function:-

- Hardware component inside computers/laptops.
- Enables devices to connect to a network.
- Provide MAC address for device identification.

7. End Devices (PCs, laptops, Printers)

Function:-

- PCs/laptops are used by users to access network resources.
- Printers are shared devices used for printing over the network.

3. Analyze Connection Type:-

- Specify whether connections are:-
 - wired (Ethernet)
 - wireless (Wi-Fi)
- Mention where each type is used.

In computers networks connections are mainly divided into wired (Ethernet) and wireless (Wi-Fi). They differ in how data is transmitted and where they are commonly used.

Wired Connection (Ethernet):-

A wire connection used physical cables (usually Ethernet cables) to connect devices directly to router switch or modem.

Where it is used:-

- Office network - desktops connected to LAN for stable communication.
- School/college - computer labs for reliable internet access.
- Data center - servers connected for high speed low latency performance.
- Gaming setups - for minimal lag and stable online games.
- Home desktops/Smart TV - when stable streaming or high speed is needed.

Key features:-

- very stable connection
- faster and lower latency
- less interference
- Requires physical cabling

Wireless Connection (Wi-Fi)

A wireless connection uses radio waves to connect devices to a router without physical cables.

Where it is used:-

- Home - smartphones, laptops, smart devices
- Public Place - cafes, airports, mall (Public hotspots)
- Campuses - student Wi-Fi across buildings
- Hotels - guest internet access.

Key features:-

No cable needed

Easy mobility and convenience

Slightly less stable than wired connection.

Can be affected by distance, walls and interference

6. Explain observation

- Briefly explain
- Difference between wired and wireless speed
- Role of Subnet Mask in networking

Wired Network

faster and more stable

Less Interference

Use Ethernet cable

More secure

Low latency

wireless Network

speed may fluctuate

More interference

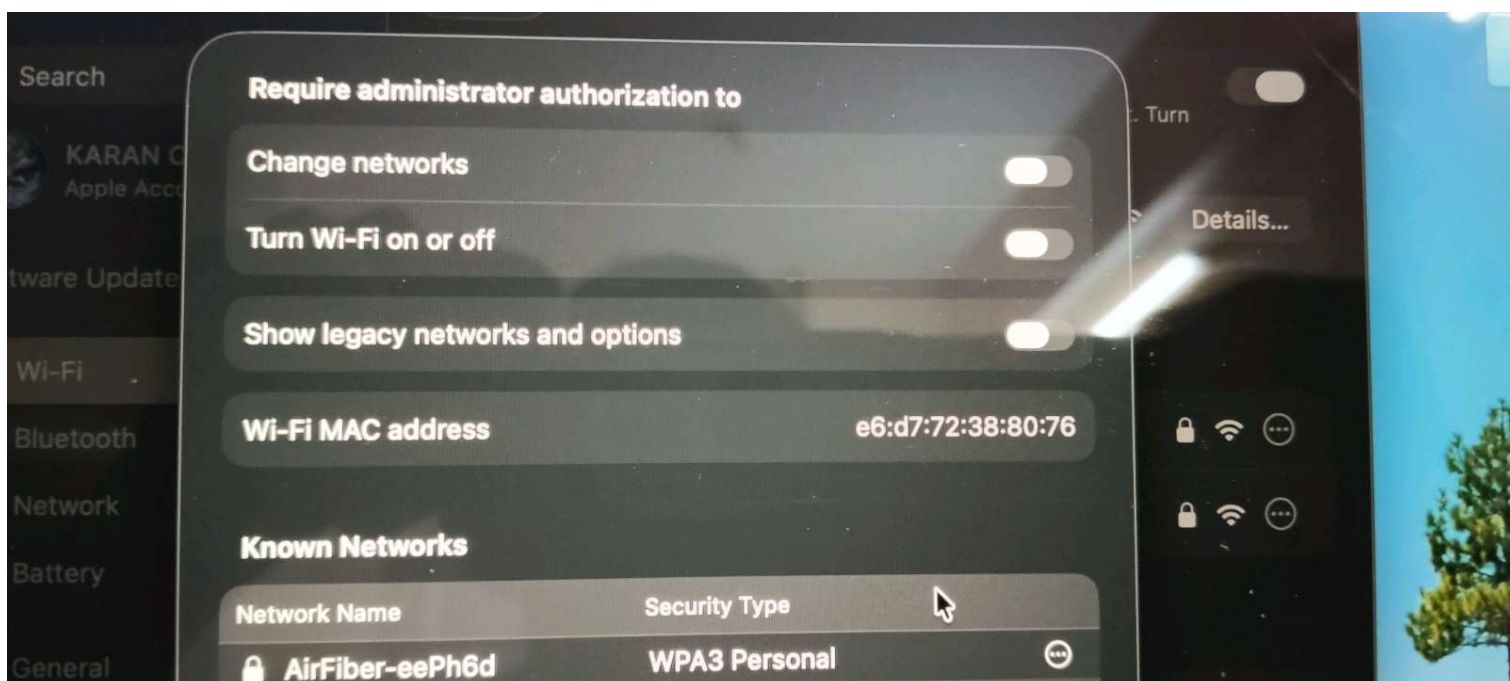
uses radiowave

comparatively less
Secure

Higher latency

Role of Subnet Mask in networking :-

The Subnet mask is used to divide an IP address into network and host portions. It helps devices determine whether another device is within the same network or outside the network. It also improves network management and reduces unnecessary traffic.



5 6 7 8 9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31

Mostly Cloudy
H:30° L:25°

karankumar — zsh — 80x7

Last login: Tue Jul 21 19:32:54 on console

[karankumar@Karan-Chauhan ~ % ipconfig getifaddr en0
10.140.88.94

[karankumar@Karan-Chauhan ~ % ipconfig getifaddr en0
10.140.88.94

karankumar@Karan-Chauhan ~ %

finished processing

[illegible]


```
Last login: Tue Jul 21 21:54:07 on ttys000
[karankumar@Karan-Chauhan ~ % ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
    options=1203<RXCSUM,TXCSUM,TXSTATUS,SW_TIMESTAMP>
    inet 127.0.0.1 netmask 0xff000000
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
    nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anpi1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether e6:4f:db:db:83:74
    media: none
    status: inactive
anpi0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether e6:4f:db:db:83:73
    media: none
    status: inactive
en3: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether e6:4f:db:db:83:53
    nd6 options=201<PERFORMNUD,DAD>
```



```
nd6 options=201<PERFORMNUD,DAD>
utun1: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
    inet6 fe80::6113:a516:4729:ce4b%utun1 prefixlen 64 scopeid 0x10
    nd6 options=201<PERFORMNUD,DAD>
utun2: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 2000
    inet6 fe80::ba39:5668:48c6:4335%utun2 prefixlen 64 scopeid 0x11
    nd6 options=201<PERFORMNUD,DAD>
utun3: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1000
    inet6 fe80::ce81:b1c:bd2c:69e%utun3 prefixlen 64 scopeid 0x12
    nd6 options=201<PERFORMNUD,DAD>
[karankumar@Karan-Chauhan ~ % networkQuality
==== SUMMARY ====
Uplink capacity: 26.293 Mbps
Downlink capacity: 13.018 Mbps
Responsiveness: Low (4.993 seconds | 12 RPM)
Idle Latency: 98.849 milliseconds | 606 RPM
karankumar@Karan-Chauhan ~ %
```